Linux System Updates and Automation

College of Technology

ITT-221

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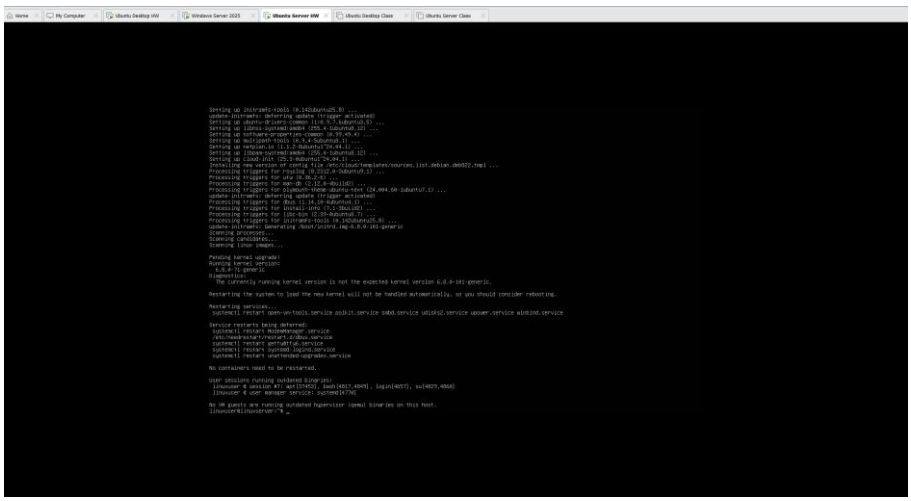
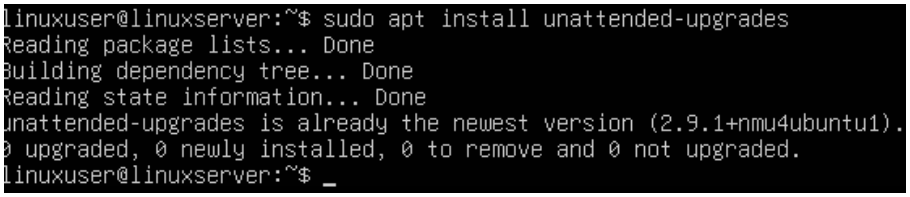
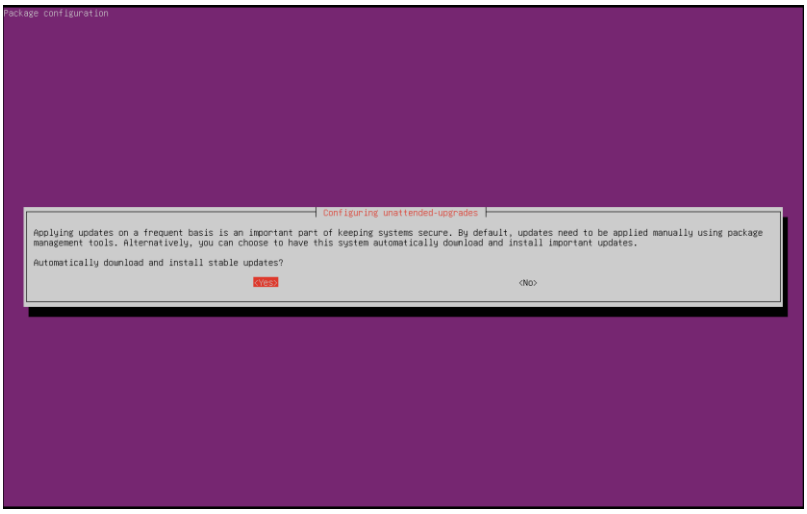
	Software update confirmation on the Linux desktop
	sudo apt upgrade completion on the Linux Server
	Installing unattended-upgrades for the Linux Server



Enabling automatic upgrades via configuration with sudo dpkg-reconfigure unattended-upgrades

```
linuxuser@linuxserver:~$ systemctl list-timers | grep apt
Sat 2026-03-07 06:13:25 UTC      9h Fri 2026-03-06 20:24:49 UTC    31min ago apt-daily-upgrade.timer        apt-daily-upgrade.service
Sat 2026-03-07 16:58:50 UTC     20h Fri 2026-03-06 20:24:49 UTC    31min ago apt-daily.timer                apt-daily.service
linuxuser@linuxserver:~$ _
```

Checking that automatic upgrades are scheduled via the existing system timers

```
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-backports_main_i18n_Tran
slation-en'  a=noble-backports,c=main,v=24.04,o=Ubuntu,l=Ubuntu arch='' site='us.archive.ubuntu.com' IndexType='Debian Translation Index' Size=121056 ID:17>
Applying pinning: PkgFilePin(id=16, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-backports_main_binary-am
d64_Packages'  a=noble-backports,c=main,v=24.04,o=Ubuntu,l=Ubuntu arch='amd64' site='us.archive.ubuntu.com' IndexType='Debian Package Index' Size=254095 ID:16>
Applying pinning: PkgFilePin(id=15, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_multiverse_i18n_
Translation-en'  a=noble-updates,c=multiverse,v=24.04,o=Ubuntu,l=Ubuntu arch='' site='us.archive.ubuntu.com' IndexType='Debian Translation Index' Size=69131 ID:
15>
Applying pinning: PkgFilePin(id=14, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_multiverse_binar
y-amd64_Packages'  a=noble-updates,c=multiverse,v=24.04,o=Ubuntu,l=Ubuntu arch='amd64' site='us.archive.ubuntu.com' IndexType='Debian Package Index' Size=161038
 ID:14>
Applying pinning: PkgFilePin(id=13, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_universe_i18n_Tr
anslation-en'  a=noble-updates,c=universe,v=24.04,o=Ubuntu,l=Ubuntu arch='' site='us.archive.ubuntu.com' IndexType='Debian Translation Index' Size=3298218 ID:13
>
Applying pinning: PkgFilePin(id=12, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_universe_binary-
amd64_Packages'  a=noble-updates,c=universe,v=24.04,o=Ubuntu,l=Ubuntu arch='amd64' site='us.archive.ubuntu.com' IndexType='Debian Package Index' Size=9809387 ID
:12>
Applying pinning: PkgFilePin(id=11, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_restricted_i18n_
Translation-en'  a=noble-updates,c=restricted,v=24.04,o=Ubuntu,l=Ubuntu arch='' site='us.archive.ubuntu.com' IndexType='Debian Translation Index' Size=12207299
ID:11>
Applying pinning: PkgFilePin(id=10, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_restricted_binar
y-amd64_Packages'  a=noble-updates,c=restrICted,v=24.04,o=Ubuntu,l=Ubuntu arch='amd64' site='us.archive.ubuntu.com' IndexType='Debian Package Index' Size=165424
42 ID:10>
Applying pinning: PkgFilePin(id=9, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_main_i18n_Transl
ation-en'  a=noble-updates,c=main,v=24.04,o=Ubuntu,l=Ubuntu arch='' site='us.archive.ubuntu.com' IndexType='Debian Translation Index' Size=6864378 ID:9>
Applying pinning: PkgFilePin(id=8, priority=-32768)
Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/us.archive.ubuntu.com_ubuntu_dists_noble-updates_main_binary-amd6
4_Packages'  a=noble-updates,c=main,v=24.04,o=Ubuntu,l=Ubuntu arch='amd64' site='us.archive.ubuntu.com' IndexType='Debian Package Index' Size=10064119 ID:8>
Using ("linux-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^kfreebsd-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^gnumach-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+
(-.+)?$|^.*-modules-[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^.*-kernel-[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^linux-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(
-.+)?$|^kfreebsd-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^gnumach-.*=[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$|^.*-modules-[1-9][0-9]*\.[0-9]+\.[0-9
]+(-.+)?$|^.*-kernel-[1-9][0-9]*\.[0-9]+\.[0-9]+-[0-9]+(-.+)?$) regexp to find kernel packages
Using ("linux-.*=6\.8\.0\-71\-generic$|^linux-.*=6\.8\.0\-71$|^kfreebsd-.*=6\.8\.0\-71\-generic$|^kfreebsd-.*=6\.8\.0\-71$|^gnumach-.*=6\.8\.0\-71\-generic$|^gn
umach-.*=6\.8\.0\-71$|^.*-modules-6\.8\.0\-71\-generic$|^.*-modules-6\.8\.0\-71$|^.*-kernel-6\.8\.0\-71\-generic$|^.*-kernel-6\.8\.0\-71$|^linux-.*=6\.8\.0\-71\
-generic$|^linux-.*=6\.8\.0\-71$|^kfreebsd-.*=6\.8\.0\-71\-generic$|^kfreebsd-.*=6\.8\.0\-71$|^gnumach-.*=6\.8\.0\-71\-generic$|^gnumach-.*=6\.8\.0\-71$|^.*-mod
ules-6\.8\.0\-71\-generic$|^.*-modules-6\.8\.0\-71$|^.*-kernel-6\.8\.0\-71\-generic$|^.*-kernel-6\.8\.0\-71$) regexp to find running kernel packages
pkgs that look like they should be upgraded:
Fetched 0 B in 0s (0 B/s)
fetch.run() result: 0
Packages blacklist due to conffile prompts: []
No packages found that can be upgraded unattended and no pending auto-removals
The list of kept packages can't be calculated in dry-run mode.
linuxuser@linuxserver:~$ _
```

Proof of the unattended-upgrades works via sudo unattended-upgrade --dry-run --debug

1) Why should a system administrator review the list of packages after apt update is run?

A system administrator should review the list to see which packages are being updated and identify any potential conflicts or risky upgrades. Not all updates are equal and some may affect critical services or dependencies, so reviewing the list helps prevent unexpected downtime. For example, updating a core package like systemd without checking could break logins or network services. According to Ubuntu documentation, reviewing package updates ensures both security and stability before installation (Ubuntu, 2026).

2) As a Linux system administrator managing patching, what is a key risk to evaluate during implementation and a single strategy to mitigate it?

A key risk is that updates can break critical system services or applications, potentially locking users out or causing downtime. The best way to avoid this is to test updates in a staging environment or take a VM snapshot before applying them to production systems. This allows administrators to roll back safely if something goes wrong.

References

Automatic updates. Ubuntu Server. (2025, December 15). https://ubuntu.com/server/docs/how-to/software/automatic-updates/